YASMEEN

Patiala, Punjab
+91-9915406994 yasmeenn00786@email.com LinkedIn/Yasmeen github.com/yasmeen-086

PROFESSIONAL SUMMARY

Robotics & AI undergraduate with hands-on experience developing **machine learning, deep learning, LLM, and RAG-based applications**. Skilled in **Python, TensorFlow, LangChain, Pandas, NumPy, model training, evaluation, data preprocessing, feature engineering, and API development**, with experience building AI applications, evaluating ML/RAG systems, and deploying production-ready inference workflows.

EDUCATION

Thapar Institute of Engineering and Technology Patiala, Punjab
B.E. Robotics & Artificial Intelligence CGPA: 8.28/10 *Aug 2023 – Present*

EXPERIENCE

RESEARCH INTERN Jun 2026 – Present
Thapar Institute of Engineering and Technology *Patiala, Punjab*

- Built a multimodal AI platform integrating **1,205 maternal clinical records** and fetal brain ultrasound images for binary pregnancy risk prediction.
- Developed and evaluated **ResNet-18 and Gradient Boosting models**, achieving **99.0% Accuracy, 0.9996 AUC, and 97.7% Recall** for binary risk prediction.
- Applied patient-level train/test splitting to prevent data leakage and used **SMOTE** and augmentation to address class imbalance.

DELOITTE AUSTRALIA – DATA ANALYTICS JOB SIMULATION Completed
Forage *Virtual*

- Completed a data analytics simulation involving **data analysis and forensic technology**, working with structured datasets to identify relevant findings; created an interactive **Tableau dashboard** to visualize data and communicate key findings.
- Used **Excel** to classify data, analyze patterns, and draw business conclusions from the available dataset.

PROJECTS

AI-Powered Pregnancy Risk Prediction Platform **(IEEE InCoWoCo 2026)** *Python, TensorFlow, React, Flask*

- Built a multimodal deep learning application combining maternal clinical data and fetal ultrasound images for pregnancy risk prediction; achieved **99.0% Accuracy and 0.9996 AUC** using Gradient Boosting and deep learning models.
- Developed secure prediction workflows with automated inference through a production-ready **React + Flask** application.

PayPulse – Incentive Compensation Analytics Platform *Python, SQL, FastAPI, Streamlit*

- Replaced a manual spreadsheet payout process with a **SQL-backed analytics platform handling 1,800+ records**, automating **KPI analysis, anomaly detection, and reporting** through stakeholder-facing interactive dashboards.
- Performed data analysis using **SQL and Python** to identify trends and generate actionable business insights.

Agentic RAG Evaluation Framework *Python, LangChain, LLM*

- Developed an automated evaluation framework for Retrieval-Augmented Generation (RAG) pipelines and evaluated retrieval relevance, answer faithfulness, and LLM response quality using structured benchmarks.
- Applied **prompt engineering** techniques to improve response quality and workflow reliability.

TECHNICAL SKILLS

Programming: Python, SQL, Data Structures & Algorithms
AI & LLM: LangChain, LLMs, RAG, Prompt Engineering, AI Applications
Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Clustering, Gradient Boosting, Random Forest, Model Training, Model Evaluation, Validation
Deep Learning: TensorFlow, ResNet-18, Data Augmentation, SMOTE, Feature Engineering
Data: Pandas, NumPy, Data Preprocessing, Data Cleaning, Exploratory Data Analysis, Data Validation, Statistical Analysis
Development: FastAPI, Flask, React, Streamlit, REST APIs, Git, GitHub, Jupyter Notebook
Databases: MySQL, Relational Database Design, Data Modeling, SQL Queries
Professional: Problem Solving, Analytical Thinking, Research, Communication, Teamwork

ACHIEVEMENTS

- Research paper accepted to the **2026 International Conference for Women in Computing**.
- Selected as an **Open Source Contributor** for GirlScript Summer of Code (GSSoC) 2026.
- Completed the **Forage AI in Action Job Simulation** focused on Prompt Engineering and AI workflow.